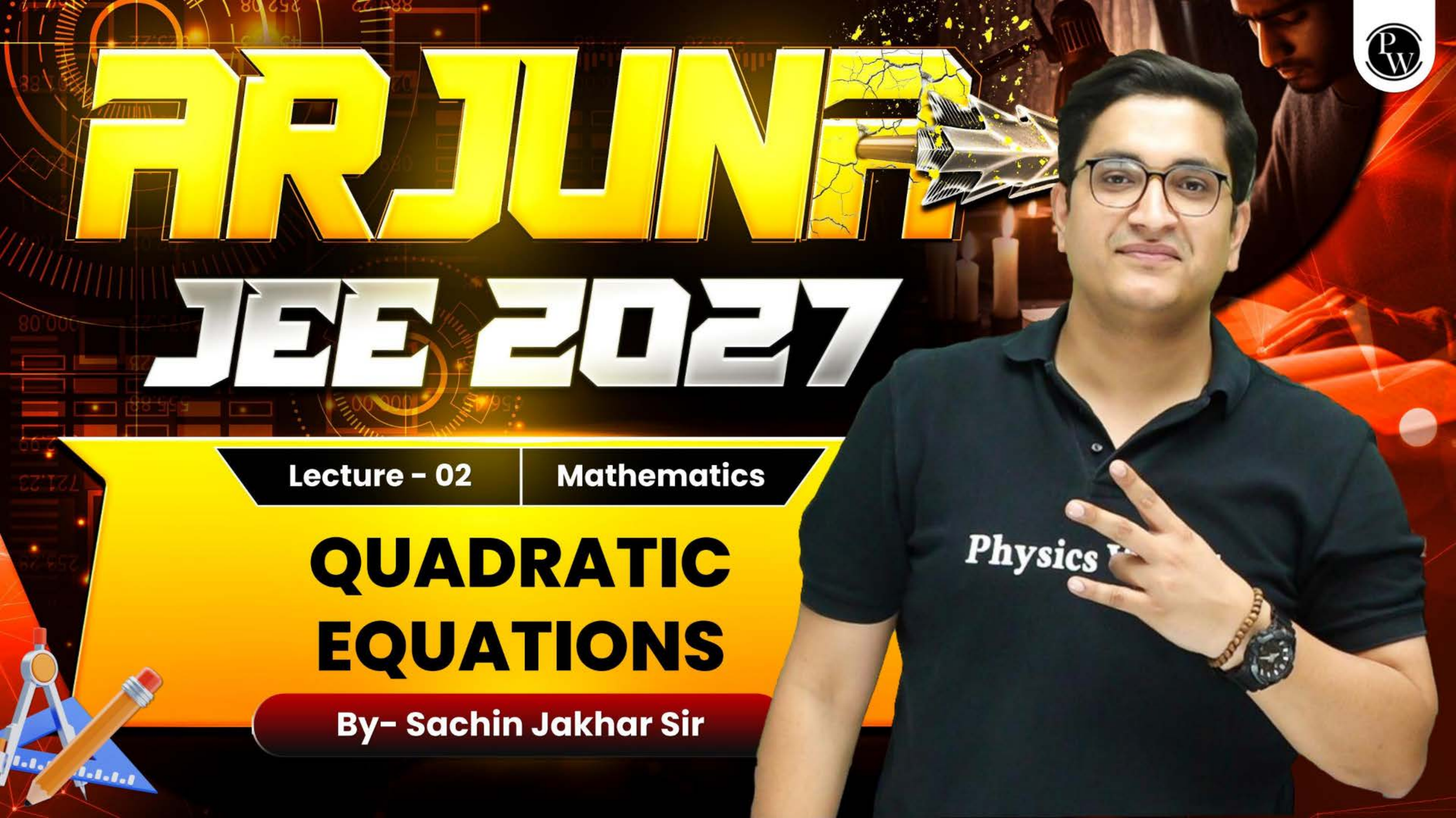
ARJUNA JEE 2027

Lecture - 02 | Mathematics

QUADRATIC EQUATIONS

By- Sachin Jakhar Sir



Topics *to be covered*

A **Relation between Roots & Coefficients**

B **Important Points about Quadratic**

C **Question Practice**

D





LAST CLASS RECAP

$$ax^2 + bx + c = 0, \quad (a \neq 0)$$

α, β

(i) Sum of Roots = $-\dfrac{\text{coefficient of } x}{\text{coefficient of } x^2} \Rightarrow \alpha + \beta = -\dfrac{b}{a}$

(ii) Product of Roots = $+\dfrac{\text{constant term}}{\text{coefficient of } x^2} \Rightarrow \alpha\beta = +\dfrac{c}{a}$

(iii) Difference of Roots = $|\alpha - \beta| = \left|\dfrac{\sqrt{D}}{a}\right|$

Note : *To find sum, product & difference of roots, roots calculate krne ki jarurat nhi hai.*

Identity

(i) $(\alpha+\beta)^2 = (\alpha-\beta)^2 + 4\alpha\beta$

(ii) $(\alpha-\beta)^2 = (\alpha+\beta)^2 - 4\alpha\beta$



$$(\alpha-\beta)^2 = \left(\frac{-b}{a}\right)^2 - 4\frac{c}{a}$$

$$= \frac{b^2}{a^2} - \frac{4c}{a}$$

$$\sqrt{(\alpha-\beta)^2} = \sqrt{\frac{b^2-4ac}{a^2}} \Rightarrow \boxed{|\alpha-\beta| = \frac{\sqrt{D}}{|a|}}$$

QUESTION PRACTICE

Module Question

x = 0, 8, 3

83%

Solve the equation $\frac{3x^4 + x^2 - 2x - 3}{3x^4 - x^2 + 2x + 3} = \frac{5x^4 + 2x^2 - 7x + 3}{5x^4 - 2x^2 + 7x - 3}$

c+d

$\frac{x}{y} = \frac{p}{q}$

c+d

$\frac{x+y}{x-y} = \frac{p+q}{p-q}$

$\frac{3x^4+x^2-2x-3+3x^4-x^2+2x+3}{3x^4+x^2-2x-3-(3x^4-x^2+2x+3)} = \frac{5x^4+2x^2-7x+3+5x^4-2x^2+7x-3}{5x^4+2x^2-7x+3-(5x^4-2x^2+7x-3)}$

$\frac{6x^4}{2x^2-4x-6} = \frac{10x^4}{4x^2-14x+6}$

A. $x = 5, 2$

B. $x = 4, 1$

C. $x = 3, 8$

D. $x = 1, 5$



$$\frac{6x^4}{2x^2-4x-6} = \frac{10x^4}{4x^2-14x+6}$$

$$\frac{6x^4}{2x^2-4x-6} - \frac{10x^4}{4x^2-14x+6} = 0$$

$$\frac{2x^4\left\{3(4x^2-14x+6) - 5(2x^2-4x-6)\right\}}{(2x^2-4x-6)(4x^2-14x+6)} = 0$$

$12x^2-42x+18-10x^2+20x+30$

$$2x^4\left(2x^2-22x+48\right) = 0$$

$$2\times 2x^4\left(x^2-11x+24\right) = 0$$

$$2\times 2x^4\left(x^2-11x+24\right) = 0$$

$$4x^4\,(x-8)(x-3) = 0$$

$x=8$, $x=3$, $x=0$

QUADRATIC EQUATIONS

Question

• 60%.

If the roots of the equation $x^2 - bx + c = 0$ be two consecutive integers, then $b^2 - 4c$ equals

A. −2

B. 3

C. 2

D. 1

diff of Roots = 1.

$\left|\frac{\sqrt{D}}{a}\right| = 1.$

$\left|\frac{\sqrt{b^2-4\times1\times c}}{1}\right| = 1.$

$\sqrt{b^2-4c} = 1$

$b^2-4c = 1.$

$n, n+1$

OR

$n-1, n$

|diff| = 1

Question [JEE Mains-2022]

The "minimum" value of the sum of the squares of the roots of $x^2 + (3-a)x + 1 = 2a$ is

A. 4

B. 5

C. 6

D. 8

$$x^2 + (3-a)x + (1-2a) = 0$$

roots: α, β

$$\alpha + \beta = -\frac{(3-a)}{1} = a-3$$

$$\alpha\beta = \frac{1-2a}{1} = 1-2a$$

A: $(\alpha+\beta)^2$ B: $(\alpha^2+\beta^2)$

$$(\alpha+\beta)^2 = \alpha^2+\beta^2+2\alpha\beta$$

??

$$(a-3)^2 = \alpha^2+\beta^2+2-4a$$

$$a^2+9-6a+4a-2 = \alpha^2+\beta^2$$

$$a^2-2a+1+6 = \alpha^2+\beta^2$$

$$(a-1)^2+6 = \alpha^2+\beta^2$$

$$\alpha^2+\beta^2 = 6+\underbrace{(a-1)^2}_{\geq 0} \Big| = 6+0 = \boxed{6}$$

min. value

Note :-

Sum of two prime no. is odd then one prime no. should be 2

{ 2, 3, 5, 7, 11, 13, ... }

2 + ☐ = odd

Acidic Question

Let $f(x) = x^2 - ax + b$ where 'a' is odd positive integer and roots of equation $f(x) = 0$ are two distinct prime numbers. If $a + b = 35$, then value of $f(10)$ = _____.

Answer: -8

different

$f(x) = x^2 - ax + b$

$x^2 - ax + b = 0$

$P_1 = 2$

$P_2 = 11$

$P_1 + P_2 = \frac{-(-a)}{1} = a$

$P_1 + P_2 = a$ (a: odd) $\Rightarrow P_1 = 2$

$f(x) = x^2 - 13x + 22$

$x = 10$

$f(10) = 10^2 - 13 \times 10 + 22 = -30 + 22 = -8$.

$x = 2$ is a root

$4 - 2a + b = 0$

$-2a + b = -4$

$a + b = 35$

$-3a = -39$

$a = 13$, $b = 35 - 13 = 22$.

$V_1^{V_2} = 1$

0^0 = not def.

ex: Solve: $(x)^{x^2-x} = 1$

- $x = 1 \rightarrow$ RHS $= 1^0 = 1$. ✓
- $x = -1 \rightarrow$ RHS $= (-1)^2 = 1$. ✓
- $x = 0 \rightarrow$ RHS $= 0^0 =$ nd. ✗

$\log_{10}(\ )$

$$\log_{10}(x)^{x^2-x} = \log_{10}(1)$$

$$(x^2-x)\left(\log_{10}|x|\right) = 0$$

$$x(x-1)\left(\log_{10}|x|\right) = 0$$

$x = 0$, $x = 1$, $x = \pm 1$.

$\log_{10}|x| = 0$

$|x| = 10^0 = 1$

$x = \pm 1$.

Critical Thinking Question (CTQ) [JEE Mains-2016]

The sum of all real values of x satisfying $(x^2 - 5x + 5)^{x^2+4x-60} = 1$ is

A 5

B 3

C −4

D 6

$\# \quad x^2-5x+4=0$

$(x-1)(x-4)=0$

$\# \quad x^2-5x+6=0$

$(x-2)(x-3)=0$

$\# \quad \log_{10}(\quad)$

$\log_{10}(x^2-5x+5)^{x^2+4x-60} = \log_{10} 1$

$(x^2+4x-60)\left(\log_{10}|x^2-5x+5|\right) = 0$

$(x+10)(x-6)$

$|x^2-5x+5| = 1$

$+ \rightarrow x^2-5x+5=1 \Rightarrow x^2-5x+4=0$

$- \rightarrow x^2-5x+5=-1 \Rightarrow x^2-5x+6=0$

Critical Thinking Question (CTQ) [JEE Mains-2016]

38%

The sum of all real values of x satisfying $(x^2 - 5x + 5)^{x^2+4x-60} = 1$ is

A 5

B 3

C −4

D 6

$(x-1)(x-4) = 0 \longrightarrow x = 1, 4$

$(x-2)(x-3) = 0$ ✗

$\hookrightarrow x = 2, 3$

$(x+10)(x-6) = 0$

$\hookrightarrow x = 6, -10$

Sum = 6.

Sum = 3.

Check:-

✓ $x = 1 \rightarrow RHS = (1)^{-55} = 1.$

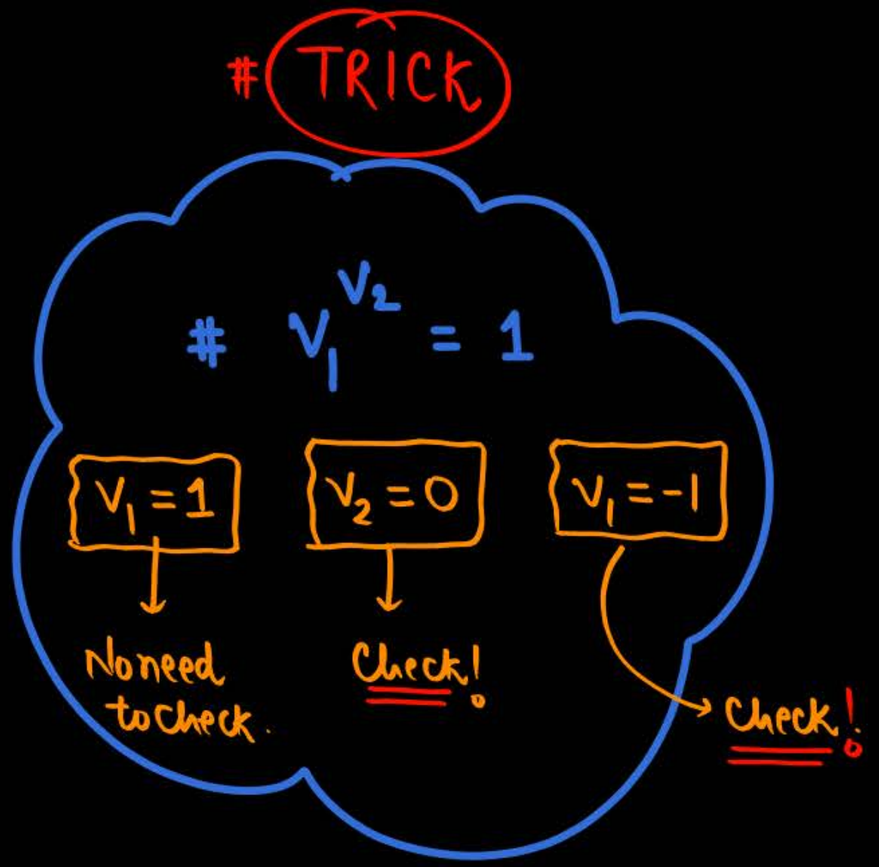
✓ $x = 4 \rightarrow RHS = (1)^{\text{even}} = 1.$

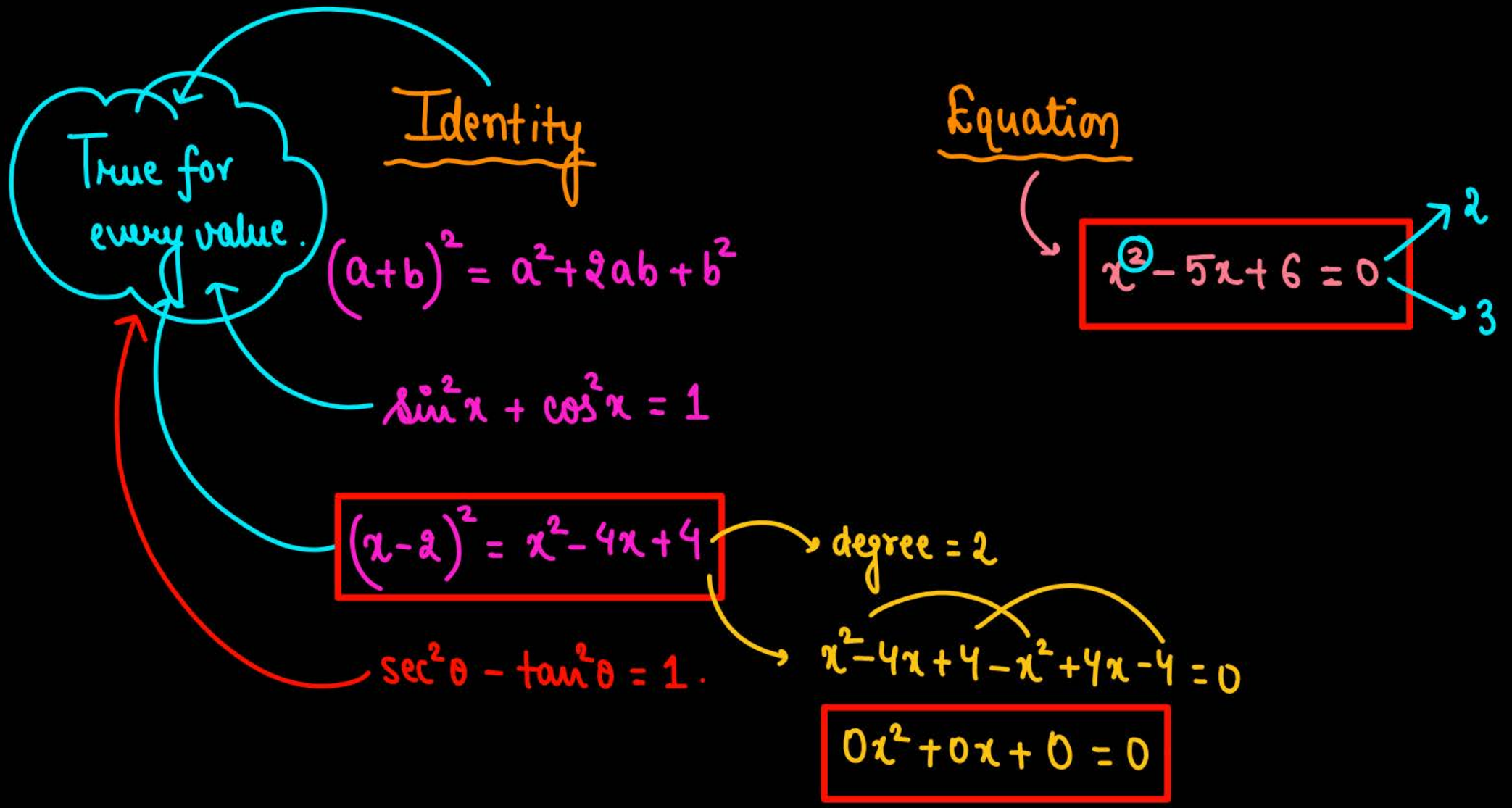
✓ $x = 2 \rightarrow RHS = (-1)^{-48} = \frac{1}{(-1)^{48}} = 1.$

✗ $x = 3 \rightarrow RHS = (-1)^{-39} = \frac{1}{(-1)^{39}} = -1$

✓ $x = 6 \rightarrow RHS = 11^0 = 1.$

✓ $x = -10 \rightarrow RHS = 155^0 = 1.$


TRICK
$v_1^{v_2} = 1$
$v_1 = 1$
$v_2 = 0$
$v_1 = -1$
No need to check.
Check!
Check!


True for every value.
Identity
$(a+b)^2 = a^2+2ab+b^2$
$\sin^2 x + \cos^2 x = 1$
$(x-2)^2 = x^2-4x+4$
$\sec^2\theta - \tan^2\theta = 1$.
Equation
$x^2 - 5x + 6 = 0$
2
3
degree = 2
$x^2-4x+4-x^2+4x-4 = 0$
$0x^2+0x+0 = 0$



EQUATION & IDENTITIES

Which is true for every value of 'x' (wherever defined).

OR

Which has more roots then its degree.

OR

In which coefficient of each term is zero.

→ for polynomial identities.

Question [AIEEE-2011]

The values of λ for which $(\lambda^2 - 3\lambda + 2)x^2 + (\lambda^2 - 5\lambda + 6)x + \lambda^2 - 4 = 0$ is an identity in x is

A 1

B 2

C −2

D 0

$(\lambda^2 - 3\lambda + 2) = 0$ $(\lambda^2 - 5\lambda + 6) = 0$ $\lambda^2 - 4 = 0$

$(\lambda-1)(\lambda-2) = 0$ $(\lambda-2)(\lambda-3) = 0$ $(\lambda-2)(\lambda+2) = 0$

$\lambda = 1, 2$ $\lambda = 2, 3$ $\lambda = 2, -2$

Question

Find the value of m such that
$(m^2 - 3m + 2)x^2 + (m^2 - 5m + 4)x + (m^2 - 7m + 6) = 0$ is an identify.

A 1

B 2

C −2

D −1

DIBY - (X)

Chat!!



IMPORTANT NOTES

Note-01

Factorisation of Quadratic:

$$ax^2+bx+c = a(x-\alpha)(x-\beta)$$ → "identity"

$$-\alpha-\beta = \frac{b}{a}$$

$$\alpha\beta = \frac{c}{a}$$

$$RHS = a\left\{x^2+\frac{b}{a}x+\frac{c}{a}\right\}$$

$$= a\left(x^2+(-\alpha-\beta)x+\alpha\beta\right)$$

$$= a\left(x^2-\alpha x-\beta x+\alpha\beta\right)$$

$$= a\left(x(x-\alpha)-\beta(x-\alpha)\right)$$

RAPID FIRE ROUND

Factorise following Quadratic and convert into Identity:

(i) $x^2 - 11x + 28 = 1(x-7)(x-4)$

(ii) $6x^2 - 5x + 1 = 6\left(x - \frac{1}{2}\right)\left(x - \frac{1}{3}\right) = (2x-1)(3x-1)$

(iii) $3x^2 - 2x - 7 = 3\left(x - \left(\frac{1+\sqrt{22}}{3}\right)\right)\left(x - \left(\frac{1-\sqrt{22}}{3}\right)\right)$

$$x = \frac{2 \pm \sqrt{4 - 4\times3\times(-7)}}{6} = \frac{2 \pm 2\sqrt{22}}{6} = \frac{1 \pm \sqrt{22}}{3}$$

Ques:- $x^2 - 3ax + 4b = 0$ $\begin{matrix}\nearrow \alpha \\ \searrow \beta\end{matrix}$ and $2|a = 4b$, $(7-\alpha)\cdot(7-\beta) = 49$.

Soln:

$$x^2 - 3ax + 4b = 1(x-\alpha)(x-\beta) \quad \text{(identity)}$$

$x = 7$.

$$49 - 21a + 4b = (7-\alpha)(7-\beta)$$

M-2

$$49 - 7\alpha - 7\beta + \alpha\beta$$

$$49 - 7(\alpha+\beta) + \alpha\beta$$

$$49 - 7(+3a) + 4b$$

$$49 - 21a + 4b$$

IMPORTANT NOTES

Note-02

Forming Quadratic Equation & Expression if Roots are given:

Sum P.O.R.

$$x^2 - (\alpha+\beta)x + (\alpha\beta) = 0$$

OR

$$x^2 - (\text{Sum})x + (\text{Prod.}) = 0$$

$$y = k\left(x^2 - (\alpha+\beta)x + (\alpha\beta)\right)$$

k → non-zero constant.

RAPID FIRE ROUND

Form Quadratic equation & expression whose roots are:

(i) -3 & 8

Sum $= -3 + 8 = 5$

Prod. $= -3 \times 8 = -24$

$x^2 - (5)x + (-24) = 0$

$x^2 - 5x - 24 = 0$

$y = k(x^2 - 5x - 24)$

(ii) $4 - \sqrt{3}$ & $4 + \sqrt{3}$

(2) Sum $= 4 - \sqrt{3} + 4 + \sqrt{3}$

$= 8$

Prod. $= (4 - \sqrt{3}) \times (4 + \sqrt{3})$

$= 16 - 3 = 13$

$x^2 - (8)x + (13) = 0$

$y = k(x^2 - 8x + 13)$

(iii) $\sqrt{2} + 1$ & 3

$x^2 - (4 + \sqrt{2})x + (3 + 3\sqrt{2}) = 0$

#

$$x^2 - 2ax + (a^2 - b^2) = 0$$

$(a-b)(a+b)$

$$x^2 - \left((a-b)+(a+b)\right)x + (a-b)(a+b) = 0$$

$(a-b)+(a+b) = 2a$

$a-b$

$a+b$.

$$x^2 - 2ax + a^2 - b^2 = 0$$

$$(x-a)^2 - (b)^2 = 0$$

$$(x-a+b)(x-a-b) = 0$$

$x-a+b=0 \Rightarrow x = a-b$

$x-a-b=0 \Rightarrow x = a+b$.



IMPORTANT NOTES

Note-03

(i) If α & β are roots of any quadratic then they will satisfy that quadratic.

(ii) Any higher degree polynomial expression in terms of roots (α & β) can be reduces to linear or constant.

RAPID FIRE ROUND

1. If α and β are roots of $x^2 - 5x - 11 = 0$ then:

(i) equation in α is $\alpha^2 - 5\alpha - 11 = 0$.

(ii) equation in β is $\beta^2 - 5\beta - 11 = 0$.

(iii) $\alpha^3 = m\alpha + n$, value of $m + n$ = 91.

(iv) $\beta^4 = m\beta + n$, value of $m + n$ = _____.

$\alpha^2 = 5\alpha + 11$

$\alpha \times$ ($\alpha^3 = 5\alpha^2 + 11\alpha$

$= 5(5\alpha + 11) + 11\alpha$

$= 25\alpha + 55 + 11\alpha$

$\alpha^3 = 36\alpha + 55$

m n

$\beta^2 = 5\beta + 11$

$\beta \times$ ($\beta^3 = 5\beta^2 + 11\beta$

$= 5(5\beta + 11) + 11\beta$

$\beta^3 = 36\beta + 55$

$\beta^3 = 36\beta + 55$

$\beta \times$ ($\beta^4 = 36\beta^2 + 55\beta$

$= 36(5\beta + 11) + 55\beta$

$= 180\beta + 55\beta + 396$

Acidic Question

If roots of $x^2 - 2x + 5 = 0$ are α and β then form a quadratic whose roots are: (i) $\alpha^2 + 5$ & $\beta^2 + 5$

Hello

Acidic Question

If roots of $x^2 - 2x + 5 = 0$ are α and β then form a quadratic whose roots

are: (ii) $\frac{5}{2-\alpha}$ & $\frac{5}{2-\beta}$

Hw.

Acidic Question

If roots of $x^2 - 2x + 5 = 0$ are α and β then form a quadratic whose roots are: (iii) $\alpha^3 + \alpha^2 - \alpha + 22$ & $\beta^3 + 4\beta^2 - 7\beta + 35$

Ans.

Question [JEE (Adv.)-2015, Paper-2]

Let S be the set of all non-zero real numbers α such that the quadratic equation $\alpha x^2 - x + \alpha = 0$ has two distinct real roots x_1 and x_2 satisfying the inequality $|x_1 - x_2| < 1$. Which of the following intervals is (are) a subset(s) of S?

Hw.

A $\left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{5}}\right)$

B $\left(-\frac{1}{\sqrt{5}}, 0\right)$

C $\left(0, \frac{1}{\sqrt{5}}\right)$

D $\left(\frac{1}{\sqrt{5}}, \frac{1}{2}\right)$

DO IT BY YOURSELF (DIBY)

DIBY-01

The quadratic equation whose roots are $1+\sqrt{2}, 1-\sqrt{2}$ is

(A) $x^2+2x-1=0$ (B) $x^2-2x-1=0$

(C) $x^2-2x+1=0$ (D) $x^2+2x+1=0$

[Ans. : (B)]

DIBY-02

If α, β are roots of the equation $ax^2-bx-c=0$, then $\alpha^2-\alpha\beta+\beta^2$ is equal to

(A) $\frac{b^2+3ac}{a^2}$ (B) $\frac{b^2-3ac}{a^2}$ (C) $\frac{b^2+2ac}{a^2}$ (D) $\frac{b^2-2ac}{a^2}$

[Ans. : (A)]

DIBY-03

If α, β are the roots of $ax^2+bx+c=0$, then the equation whose roots are $2+\alpha$ and $2+\beta$ is

(A) $ax^2+(b-4a)x+(4a-2b+c)=0$ (B) $ax^2-(b-4a)x+(4a-2b+c)=0$

(C) $ax^2+(b-4a)x-(4a-2b+c)=0$ (D) $ax^2-(b-4a)x-(4a-2b+c)=0$

[Ans. : (A)]

DIBY-04

If α and β are roots of the equation $x^2 + px + \frac{3p}{4} = 0$, such that $|\alpha - \beta| = \sqrt{10}$, then p belongs to the set

(A) $\{2, -5\}$ (B) $\{-3, 2\}$ (C) $\{3, -5\}$ (D) $\{-2, 5\}$

[Ans. : (D)]

DIBY-05

If α, β are the roots of $ax^2 + bx + c = 0$ and $c \neq 0$ then the value of $\frac{1}{(a\alpha+b)^2} + \frac{1}{(a\beta+b)^2}$ in terms of a, b, c is

(A) $\frac{b^2-2ac}{ac}$ (B) $\frac{2ac-b^2}{ac}$ (C) $\frac{b^2-2ac}{a^2c^2}$ (D) $\frac{b^2}{2}$

[Ans. : (C)]

DIBY-06

Let α and β be the roots of the quadratic equation $(x-2)(x-3) + (x-3)(x+1) + (x+1)(x-2) = 0$. Find the value of

$$\frac{1}{(\alpha+1)(\beta+1)} + \frac{1}{(\alpha-2)(\beta-2)} + \frac{1}{(\alpha-3)(\beta-3)}$$

Ans. : 0]



Homework

Re-attempt all Class Examples

DPP & DIBY

Module: Basic Maths (4 Sittings Each of 1 Hr.)

Exercise-03(BRIDGE): COMPLETE

Exercise-04(JEE ADV. LEVEL):COMPLETE

It is never about end result

It's always about Journey

Thank You

#futureIITians